



Hospital Engineering and Management

Lecture 13: HVAC and Air-Conditioning Systems in Hospitals

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Course Code: UBM702

Learning Objectives

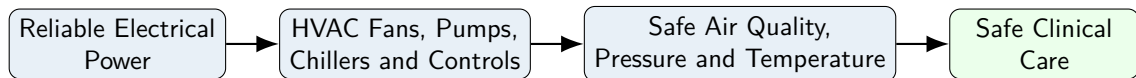
By the end of this lecture, students should be able to:

- ▶ explain why HVAC is a clinical-safety system in hospitals, not merely a comfort system;
- ▶ identify the major components of hospital air-conditioning and ventilation systems;
- ▶ calculate air changes per hour (ACH), ventilation flow and basic sensible cooling load;
- ▶ explain filtration, HEPA filtration, pressure relationships and airflow direction;
- ▶ distinguish ventilation requirements of operating rooms, ICUs and isolation/protective rooms;
- ▶ describe HVAC monitoring, maintenance, balancing, commissioning and emergency response;
- ▶ relate HVAC performance to infection control, patient safety and biomedical equipment reliability.

Connection with Lecture 12

Lecture 12 Established

A hospital needs reliable electrical power to keep life-safety and clinical systems operating continuously.



Key Link

Failure of HVAC in a critical area can become a patient-safety and infection-control emergency even when electrical power is available.

Opening Question: Is Hospital AC Just for Comfort?

Think About It

An operating theatre is cool and comfortable. Does that automatically mean its HVAC system is safe?

- ▶ What if the room is at the correct temperature but is under negative pressure?
- ▶ What if the HEPA filter is bypassing air around its frame?
- ▶ What if the required airflow is not reaching the surgical field?
- ▶ What if humidity is high enough to cause condensation or microbial growth?
- ▶ What if an isolation room exhaust fan fails silently?

Comfortable air is not necessarily safe air.

Why HVAC Is Critical in Hospitals

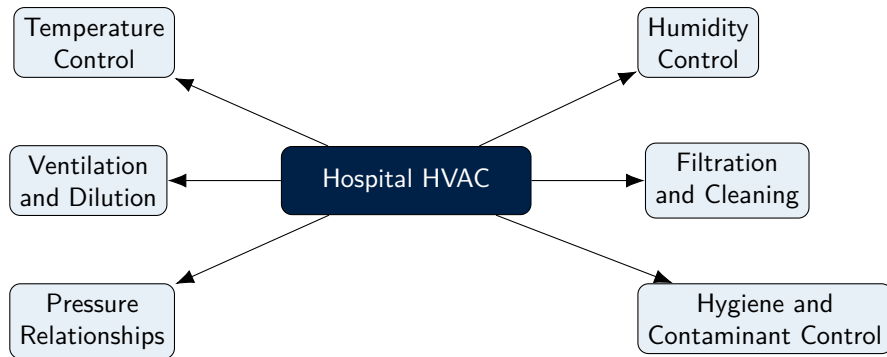
Hospital HVAC must simultaneously support:

- ▶ **thermal comfort:** acceptable temperature and humidity;
- ▶ **infection control:** dilution and removal of airborne contaminants;
- ▶ **airflow control:** movement from clean to less-clean spaces where required;
- ▶ **filtration:** removal of particles and microorganisms carried by particles;
- ▶ **odor control:** toilets, laboratories, kitchens and dirty utility areas;
- ▶ **equipment reliability:** stable environmental conditions for sensitive devices;
- ▶ **staff performance:** acceptable conditions during long procedures.

Engineering Perspective

Hospital HVAC is an environmental-control system integrated with infection prevention, clinical workflow and facility safety.

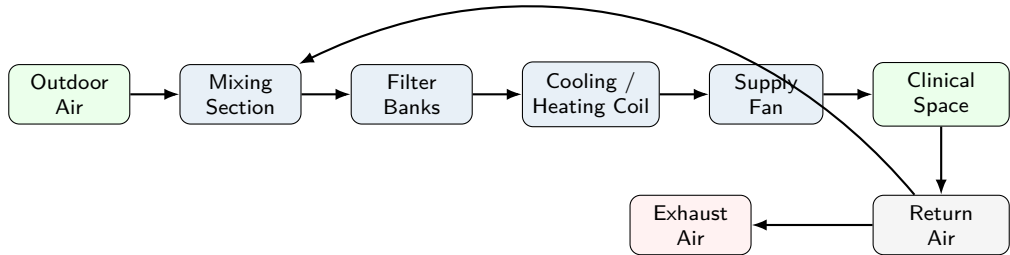
Functions of a Hospital HVAC System



Basic HVAC Terminology

- ▶ **HVAC**: Heating, Ventilation and Air Conditioning.
- ▶ **Ventilation**: intentional supply and/or removal of air from a space.
- ▶ **Outdoor air**: fresh air brought from outside for ventilation.
- ▶ **Supply air**: conditioned air delivered to the room.
- ▶ **Return air**: air taken back to the air-handling system.
- ▶ **Exhaust air**: air discharged outside and not returned to occupied spaces.
- ▶ **Recirculated air**: return air that is filtered/conditioned and reused.
- ▶ **ACH**: number of times an equivalent room-air volume is supplied or removed each hour.

Central HVAC: Simplified Air Path



Important

The actual arrangement depends on the clinical risk, required outdoor-air fraction, filtration, pressurization and whether return air may be recirculated.

Major Components of an Air-Handling Unit (AHU)

An AHU may contain:

- ▶ outdoor-air and return-air dampers;
- ▶ pre-filters and fine filters;
- ▶ cooling and/or heating coils;
- ▶ drain pan and condensate drainage;
- ▶ humidification or dehumidification components where required;
- ▶ supply fan and sometimes return/exhaust fans;
- ▶ sound attenuators;
- ▶ final filters or terminal HEPA filters in selected areas;
- ▶ sensors for temperature, humidity, pressure and filter differential pressure.

AHU Versus Room Air Conditioner

Central/AHU-Based System

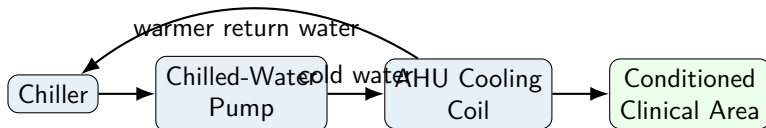
- ▶ controlled outdoor air;
- ▶ staged filtration;
- ▶ pressure control;
- ▶ centralized monitoring;
- ▶ suitable for critical areas.

Typical Room/Window AC

- ▶ mainly recirculates room air;
- ▶ limited fresh-air control;
- ▶ limited filtration;
- ▶ poor pressure-control capability;
- ▶ unsuitable as the sole solution for many critical clinical spaces.

Cooling a room and ventilating a room are different engineering tasks.

Chilled-Water Air-Conditioning System



- ▶ Common in medium and large hospitals.
- ▶ Allows multiple AHUs to serve different clinical zones.
- ▶ Plant redundancy may be provided using multiple chillers and pumps.
- ▶ Water treatment, condenser performance and pump reliability affect system efficiency.

Psychrometric Variables Engineers Must Understand

Hospital air-conditioning involves simultaneous control of:

- ▶ **dry-bulb temperature**: ordinary measured air temperature;
- ▶ **relative humidity (RH)**: moisture content relative to saturation at that temperature;
- ▶ **dew-point temperature**: temperature at which condensation begins;
- ▶ **humidity ratio**: mass of water vapor per mass of dry air;
- ▶ **enthalpy**: total heat content of moist air.

Clinical Significance

Incorrect temperature or humidity can affect patient comfort, microbial growth, condensation, static electricity, sterile supplies and some medical equipment.

Sensible and Latent Heat

Sensible Heat

Changes air temperature without changing moisture content.

- ▶ people;
- ▶ lighting;
- ▶ medical equipment;
- ▶ walls and solar gain.

Latent Heat

Associated with moisture added to or removed from air.

- ▶ occupants;
- ▶ outside humid air;
- ▶ wet processes;
- ▶ steam and water sources.

Cooling coils often have to cool *and* dehumidify simultaneously.

Air Changes per Hour (ACH)

For a room of volume V supplied with airflow Q :

$$ACH = \frac{3600Q}{V}$$

where Q is in m^3/s and V is in m^3 .
Equivalently, if airflow is in m^3/h :

$$Q_h = ACH \times V$$

Interpretation

12 ACH means the total supplied air volume in one hour is equivalent to 12 times the room volume. It does **not** mean that every air molecule is replaced exactly 12 times.

Numerical Example 1: Isolation-Room Airflow

A room measures 5 m × 4 m × 3 m and requires 12 ACH.

$$V = 5 \times 4 \times 3 = 60 \text{ m}^3$$

$$Q_h = 12 \times 60 = 720 \text{ m}^3/\text{h}$$

$$Q = \frac{720}{3600} = 0.20 \text{ m}^3/\text{s}$$

Result

The room requires approximately **720 m³/h** of equivalent ventilation airflow to achieve 12 ACH, before considering pressure-control leakage and system design margins.

What Does More ACH Do?

Increasing ACH can:

- ▶ increase dilution and removal rate of airborne contaminants;
- ▶ reduce contaminant residence time when airflow distribution is effective;
- ▶ increase fan energy and conditioning load;
- ▶ increase noise or drafts if poorly designed;
- ▶ disturb required pressure relationships if supply/exhaust balance is changed blindly.

Important

More ACH is not automatically better. Current ASHE guidance warns that simply increasing ACH can unintentionally alter room pressurization.

Airflow Direction and Pressure Relationships

Air flows from a region of higher pressure toward a region of lower pressure.



- ▶ **Positive room:** supply exceeds return/exhaust/leakage, so air tends to flow outward.
- ▶ **Negative room:** exhaust exceeds supply, so air tends to flow inward.
- ▶ **Neutral room:** no deliberate directional pressure relationship.

Positive-Pressure Rooms

Positive pressure is used when the priority is to protect the **patient or clean field** from surrounding contamination.

- ▶ operating rooms;
- ▶ selected clean rooms and sterile-processing clean zones;
- ▶ protective environments for severely immunocompromised patients;
- ▶ some clean preparation areas.

Concept

Protect what is inside the room: filtered air leaks outward when doors or small gaps are present.

Negative-Pressure Rooms

Negative pressure is used when the priority is to contain **airborne contamination generated inside** a room.

- ▶ airborne infection isolation (All) rooms;
- ▶ bronchoscopy and some aerosol-generating procedure rooms;
- ▶ selected laboratories and dirty utility areas;
- ▶ toilets and areas where odors/contaminants should not spread.

Concept

Contain what is inside the room: air enters from the corridor and contaminated room air is safely exhausted or appropriately filtered.

Pressure Differential: A Practical Reference

CDC environmental-control guidance describes a pressure difference of approximately **2.5 Pa** (0.01 in. water gauge) as a reference for directional pressure in special rooms.

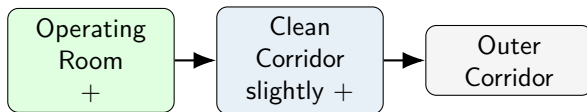
- ▶ All rooms: negative relative to adjacent spaces.
- ▶ Protective environments: positive relative to adjacent spaces.
- ▶ Doors and envelope leakage strongly influence whether pressure is maintained.
- ▶ Differential-pressure monitors provide continuous indication.

Design Caution

The exact required pressure relationship and magnitude must follow the edition of the applicable national/local standard and authority having jurisdiction (AHJ).

Pressure Cascade

A pressure cascade intentionally creates a sequence of pressure levels.



- ▶ Air is intentionally encouraged to move from cleaner to less-clean zones.
- ▶ Door opening, ceiling leaks and poor balancing can collapse the cascade.
- ▶ Pressure should be checked under realistic operating conditions.

Air Filtration: Why It Matters

Air filters remove particles that may carry dust, allergens and microorganisms.

- ▶ Coarse filters protect downstream components.
- ▶ Fine filters remove smaller particles before air enters critical zones.
- ▶ HEPA filters are used where very high particulate removal is required.
- ▶ Dirty filters increase pressure drop and fan energy.
- ▶ Poorly sealed filter frames allow **bypass**, defeating filter efficiency.

Engineering Rule

A high-efficiency filter is only as good as its installation, seal, monitoring and maintenance.

HEPA Filtration

CDC guidance describes HEPA filtration as at least **99.97% efficient for 0.3 μm particles** under the cited test convention.

- ▶ HEPA filters are commonly used in protective environments and selected high-risk clinical spaces.
- ▶ They may be installed centrally or at terminal locations depending on design.
- ▶ Differential pressure across the filter indicates loading.
- ▶ Integrity/leak testing is important in critical applications.
- ▶ Prefiltration extends HEPA filter life.

Filter Pressure Drop and Fan Energy

As filters load with dust:

- ▶ resistance to airflow increases;
- ▶ airflow may fall if fan speed/control does not compensate;
- ▶ fan energy may increase if the system maintains constant airflow;
- ▶ critical-room ACH and pressure can deteriorate;
- ▶ filter alarms may occur when differential pressure exceeds set limits.

Filter maintenance is both an infection-control and energy-management task.

Supply and Return Air Distribution

Good room ventilation depends on more than total airflow.

- ▶ Diffusers should distribute clean air without uncomfortable drafts.
- ▶ Return/exhaust grilles should help sweep contaminants away from clean zones.
- ▶ Short-circuiting between supply and return should be avoided.
- ▶ Obstructions, curtains and equipment can alter airflow patterns.
- ▶ Ceiling supply and low-level return/exhaust are common strategies in operating rooms.

Key Concept

Two rooms with identical ACH can have very different contaminant-removal performance if their airflow patterns differ.

Operating-Room HVAC: Main Objectives

The operating room HVAC system must:

- ▶ maintain positive pressure relative to adjacent areas;
- ▶ deliver filtered air to the surgical zone;
- ▶ maintain required outdoor and total airflow;
- ▶ control temperature and humidity;
- ▶ remove heat generated by lights, staff and equipment;
- ▶ minimize introduction of contaminants from corridors;
- ▶ operate reliably during scheduled procedures.

Operating-Room Ventilation: Standards Matter

Historical CDC Reference

CDC environmental-control guidance cites operating-room ventilation of approximately **15 ACH** with positive pressure in its legacy table.

Current Design Practice

Modern projects should use the currently adopted **ANSI/ASHRAE/ASHE Standard 170-2025** (or applicable national/local standard). Different jurisdictions may enforce different editions and requirements.

Do not design a hospital using a memorized number without checking the governing standard.

Why Operating-Room Doors Matter

Every unnecessary door opening can disturb the engineered environment.

- ▶ pressure differential temporarily changes;
- ▶ corridor air may enter the room;
- ▶ airflow patterns can be disturbed;
- ▶ staff traffic increases particle generation;
- ▶ temperature/humidity recovery load increases.

Management + Engineering

Maintaining OR air quality requires both correct HVAC design and disciplined clinical workflow.

Laminar / Unidirectional Airflow

Unidirectional airflow systems attempt to deliver low-turbulence, highly filtered air across the protected surgical zone.

- ▶ often use large ceiling diffuser arrays;
- ▶ recirculated air may pass through HEPA filtration;
- ▶ airflow pattern can be disrupted by lights, booms and personnel;
- ▶ installation and commissioning are critical;
- ▶ evidence for reducing overall surgical-site infection is not uniform across all procedures.

Engineering Lesson

Technology alone cannot compensate for poor aseptic practice, excessive traffic or incorrect system operation.

ICU HVAC Considerations

ICUs require reliable environmental control because patients may be highly vulnerable.

- ▶ stable temperature and humidity;
- ▶ effective filtration and ventilation;
- ▶ special isolation rooms within or near the ICU;
- ▶ minimized drafts around critically ill patients;
- ▶ continuous operation with emergency-power support for essential HVAC components;
- ▶ coordination with equipment heat loads and dense bedside technology.

Airborne Infection Isolation (All) Room

A typical All strategy includes:

- ▶ negative pressure relative to corridor;
- ▶ inward airflow through the doorway;
- ▶ controlled exhaust to a safe outdoor location or appropriately treated air where permitted;
- ▶ sealed room envelope as practical;
- ▶ continuous or readily visible pressure monitoring;
- ▶ self-closing doors where required;
- ▶ adequate air changes according to the governing standard.

All Room: CDC Reference Values

CDC guidance states:

- ▶ existing All rooms: at least 6 ACH in its cited legacy specification;
- ▶ renovated/new All rooms: at least 12 ACH;
- ▶ pressure is negative relative to adjacent spaces;
- ▶ air should generally be exhausted outdoors or treated appropriately before recirculation;
- ▶ airflow direction should be verified.

Use Correctly

These values are useful for teaching fundamentals. Actual design must follow the currently adopted standard and local regulatory requirements.

Protective Environment (PE)

A protective environment is intended to reduce exposure of severely immunocompromised patients to airborne environmental microorganisms.

- ▶ positive pressure relative to corridor;
- ▶ highly filtered supply air;
- ▶ controlled, sealed room envelope;
- ▶ sufficient ACH;
- ▶ continuous pressure indication in modern designs.

All protects others from the patient; PE protects the patient from others.

All Versus Protective Environment

Feature	All Room	Protective Environment
Primary purpose	contain airborne infection	protect vulnerable patient
Pressure	negative	positive
Air direction	corridor → room	room → corridor
Exhaust focus	prevent contaminated recirculation	maintain clean supply
Typical patient	airborne infectious disease	severe immunosuppression

Can One Room Switch Between Positive and Negative?

Older hospitals sometimes used variable-pressure rooms.

Problem

Switching pressure modes creates risk of human error, control failure and uncertainty about whether the room is truly operating as intended.

CDC guidance notes that variable-pressure rooms are discouraged for new/renovated construction.

Preferred Approach

Use dedicated room functions with engineered controls and clear pressure monitoring rather than relying on routine manual switching.

Laboratory and Diagnostic Areas

Hospital HVAC requirements vary with hazards and processes.

- ▶ microbiology laboratories may require directional airflow and controlled exhaust;
- ▶ fume-generating processes require local exhaust;
- ▶ radiology and imaging rooms may have high equipment heat loads;
- ▶ MRI suites require careful coordination with equipment cooling and emergency exhaust requirements;
- ▶ nuclear medicine and other specialized areas may have additional regulatory requirements.

Pharmacy and Sterile-Preparation Areas

Sterile-compounding and clean preparation areas require controlled air quality.

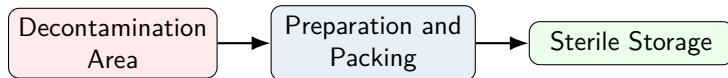
- ▶ filtration and particle control;
- ▶ pressure relationships based on type of preparation;
- ▶ appropriate temperature and humidity;
- ▶ monitoring and qualification;
- ▶ separation of clean and potentially hazardous processes.

Recall Lecture 9

Medication safety depends not only on pharmacy workflow but also on the engineering environment in which sterile products are prepared.

CSSD and HVAC Zoning

Recall Lecture 10: CSSD has physically different contamination levels.



- ▶ airflow and pressure should support dirty-to-clean separation;
- ▶ humidity control supports packaging and storage;
- ▶ ventilation must coordinate with heat and steam-generating equipment.

Temperature Control in Hospitals

Different spaces need different thermal conditions because of:

- ▶ patient age and clinical condition;
- ▶ staff clothing and activity level;
- ▶ heat generated by equipment;
- ▶ infection-control requirements;
- ▶ procedure-specific needs;
- ▶ medication, laboratory and material storage requirements.

Example

Operating rooms are often cooler than general patient rooms, while neonatal and pediatric areas may require warmer conditions.

Humidity Control

Too much humidity may:

- ▶ promote condensation and microbial growth;
- ▶ damage building finishes;
- ▶ increase discomfort.

Too little humidity may:

- ▶ increase dryness and discomfort;
- ▶ contribute to electrostatic effects in some environments;
- ▶ affect materials and processes.

The required RH range is space-specific and standards-driven.

Condensation: Hidden HVAC Risk

Condensation occurs when a surface temperature falls below the dew point of surrounding air.

- ▶ wet ducts or insulation can support microbial growth;
- ▶ condensate can damage ceilings and equipment;
- ▶ drain pans with standing water can become contamination sources;
- ▶ chilled pipes must be properly insulated and vapor-sealed.

Maintenance Clue

A ceiling stain near an AHU or chilled-water line should never be treated as only a cosmetic problem.

Basic Sensible Heat Equation

The sensible heat transferred by an air stream is approximately:

$$\dot{Q}_s = \rho c_p \dot{V} \Delta T$$

where:

- ▶ ρ = air density (approximately 1.2 kg/m³ near room conditions),
- ▶ c_p = specific heat of air (approximately 1.0 kJ/kg·K),
- ▶ $\dot{V}$ = airflow in m³/s,
- ▶ ΔT = temperature difference in °C or K.

Numerical Example 2: Sensible Cooling

Airflow to a room is $1.5 \text{ m}^3/\text{s}$. Supply air is 16°C and room air is 24°C .

$$\dot{Q}_s = 1.2 \times 1.0 \times 1.5 \times (24 - 16)$$

$$\dot{Q}_s = 14.4 \text{ kW}$$

Interpretation

The air stream removes approximately **14.4 kW of sensible heat**. The total cooling load may be higher because latent heat must also be removed.

Major Sources of Hospital Cooling Load

- ▶ outdoor air brought in for ventilation;
- ▶ solar heat gain through walls, roof and glazing;
- ▶ occupants;
- ▶ lighting;
- ▶ computers and medical devices;
- ▶ sterilizers and heat-producing clinical processes;
- ▶ infiltration through doors and leakage;
- ▶ latent moisture load from occupants and outdoor air.

Why Medical Equipment Matters to HVAC Design

Biomedical equipment can significantly alter heat loads.

- ▶ CT, MRI and angiography equipment may produce substantial heat;
- ▶ server rooms and PACS infrastructure require continuous cooling;
- ▶ multiple patient monitors and infusion devices add heat in ICUs;
- ▶ lasers and robotic systems can increase OT cooling demand;
- ▶ equipment upgrades can exceed the original HVAC design load.

Biomedical Engineer's Role

Provide accurate equipment heat-rejection, electrical and environmental requirements during planning and renovation.

HVAC and Biomedical Equipment Reliability

Poor HVAC performance may cause:

- ▶ overheating and premature component failure;
- ▶ condensation inside electronics;
- ▶ inaccurate measurements or calibration drift;
- ▶ imaging-system shutdowns;
- ▶ alarm generation and downtime;
- ▶ shortened battery life in hot technical spaces.

Environmental control is part of equipment reliability engineering.

Ventilation Versus Air Cleaning

Ventilation

Brings outdoor air and removes/dilutes indoor contaminants.

Air Cleaning

Removes particles or microorganisms from air by filtration or other treatment.

Important

Portable HEPA units can increase equivalent clean-air delivery, but they do not by themselves satisfy outdoor-air requirements.

Portable HEPA Units

Portable HEPA units may be useful when:

- ▶ existing ventilation is temporarily inadequate;
- ▶ additional air cleaning is required during an emergency;
- ▶ fixed HVAC is under repair;
- ▶ a room requires temporary enhancement while a permanent solution is developed.

Limitations

Performance depends on room layout, unit placement, airflow obstructions and delivered clean-air rate. Poor placement can create dead zones or disturb intended airflow.

Ultraviolet Germicidal Irradiation (UVGI)

UVGI can be used as an **adjunct** air-treatment method in selected applications.

- ▶ may be installed in ducts or upper-room systems;
- ▶ effectiveness depends on UV dose, exposure time and maintenance;
- ▶ dirty lamps lose effectiveness;
- ▶ safety controls are required to prevent harmful exposure;
- ▶ it does not replace required ventilation, pressure control or HEPA filtration where those are specified.

Building Management System (BMS) for HVAC

A BMS can monitor and trend:

- ▶ room and duct temperatures;
- ▶ relative humidity;
- ▶ differential pressure between critical rooms;
- ▶ filter differential pressure;
- ▶ fan and pump status;
- ▶ valve and damper positions;
- ▶ chilled-water temperatures;
- ▶ alarms and energy consumption.

Value

Trend data helps engineers detect slow deterioration before clinical staff notice a failure.

Critical HVAC Alarms

Examples of alarms requiring rapid response:

- ▶ All room loses negative pressure;
- ▶ OR loses positive pressure;
- ▶ supply or exhaust fan trips;
- ▶ room temperature or RH leaves allowable range;
- ▶ HEPA/filter differential pressure is abnormal;
- ▶ chilled-water supply temperature rises;
- ▶ condensate overflow is detected;
- ▶ smoke/fire system commands HVAC shutdown or smoke-control mode.

Emergency Power for HVAC

Not every HVAC component can be backed up equally.

- ▶ prioritize critical spaces: OR, ICU, AII/PE, transplant/oncology and essential technical rooms;
- ▶ emergency power may support selected AHUs, exhaust fans, controls and pumps;
- ▶ chiller backup strategy depends on hospital size and risk assessment;
- ▶ restart sequencing is important to avoid excessive generator loading;
- ▶ loss of ventilation must trigger clinical contingency procedures.

HVAC Redundancy

Redundancy can be provided at several levels:

- ▶ multiple chillers;
- ▶ duty/standby chilled-water pumps;
- ▶ redundant cooling-tower cells;
- ▶ standby exhaust fan for critical isolation areas;
- ▶ dual temperature sensors or pressure monitoring in high-risk areas;
- ▶ spare filters, belts and critical control components.

Design Question

What is the consequence if this component fails at 2:00 a.m. during a surgery or outbreak?

Preventive Maintenance of AHUs

Typical tasks include:

- ▶ inspect/replace filters;
- ▶ clean coils and drain pans;
- ▶ verify condensate drainage;
- ▶ inspect belts, bearings and vibration;
- ▶ check damper operation;
- ▶ inspect insulation and duct leakage;
- ▶ calibrate temperature, humidity and pressure sensors;
- ▶ verify fan airflow and motor condition.

Chiller and Pump Maintenance

Key maintenance activities:

- ▶ monitor refrigerant pressures and temperatures;
- ▶ verify chilled-water supply/return temperatures;
- ▶ check condenser performance and water treatment;
- ▶ monitor pump vibration, seals and bearings;
- ▶ inspect strainers and valves;
- ▶ track compressor current and energy consumption;
- ▶ perform seasonal and manufacturer-recommended service.

Filter Maintenance Strategy

- ▶ establish filter inventory and replacement intervals;
- ▶ trend differential pressure rather than relying only on calendar replacement;
- ▶ inspect filter frames and gaskets for bypass;
- ▶ protect staff during removal in contaminated locations;
- ▶ document date, location, type and differential pressure;
- ▶ verify critical-area airflow after major filter changes.

Air Balancing

Testing, Adjusting and Balancing (TAB) verifies that actual airflow matches design intent.

- ▶ measure supply, return and exhaust airflow;
- ▶ adjust dampers and fan settings;
- ▶ verify room pressure relationships;
- ▶ confirm ACH;
- ▶ check diffuser distribution and comfort;
- ▶ repeat after major renovations or system changes.

Commissioning Critical HVAC Spaces

Before clinical occupancy, verify:

- ▶ correct room pressure direction;
- ▶ required airflow/ACH;
- ▶ temperature and humidity control;
- ▶ filter installation and integrity where applicable;
- ▶ alarm and BMS functions;
- ▶ door effects on pressure;
- ▶ emergency-power response;
- ▶ documentation and staff training.

Hospital Renovation and Construction Risk

Construction releases dust and fungal spores and can disrupt pressure zones.

- ▶ isolate work areas;
- ▶ maintain appropriate negative pressure in construction zones;
- ▶ protect nearby high-risk patient areas;
- ▶ seal penetrations and control dust transport;
- ▶ verify HVAC performance after work completion;
- ▶ coordinate engineering, infection-control and contractor teams.

Lesson

A small ceiling-opening job near an immunocompromised unit can become an infection-control risk if airflow is not managed.

HVAC Failure Modes

Common failure modes include:

- ▶ fan motor or VFD failure;
- ▶ clogged or bypassed filters;
- ▶ chilled-water pump failure;
- ▶ chiller trip;
- ▶ sensor drift or calibration error;
- ▶ stuck damper or control valve;
- ▶ duct leakage;
- ▶ condensate blockage;
- ▶ loss of BMS communication;
- ▶ incorrect balancing after renovation.

FMEA Example: Isolation-Room Exhaust Fan

Failure	Effect	Detection	Control
Exhaust fan trip	loss of negative pressure	pressure alarm / BMS	standby fan + clinical response
Filter blockage	reduced airflow	differential pressure	PM and filter change
Door held open	pressure collapse	room monitor / observation	workflow control
Sensor drift	false safe indication	periodic calibration	independent verification

HVAC Performance Indicators

Useful engineering KPIs include:

- ▶ percentage of time critical rooms maintain required pressure;
- ▶ temperature/RH compliance rate;
- ▶ HVAC-related clinical downtime;
- ▶ number of critical alarms per month;
- ▶ PM compliance for AHUs/chillers/exhaust systems;
- ▶ filter life and differential-pressure trends;
- ▶ energy consumption per occupied bed or floor area;
- ▶ mean time to restore critical ventilation failures.

Energy Use in Hospital HVAC

HVAC is often one of the largest energy consumers in a hospital because it operates continuously and conditions large outdoor-air volumes.

- ▶ fans and pumps;
- ▶ chillers/compressors;
- ▶ cooling towers;
- ▶ reheating and humidity control;
- ▶ 24/7 ventilation in critical areas.

Challenge

Reduce energy without compromising infection control, pressure relationships or patient safety.

Energy-Efficiency Strategies

Possible measures include:

- ▶ variable-speed drives on fans and pumps;
- ▶ high-efficiency chillers and motors;
- ▶ optimized chilled-water temperature and sequencing;
- ▶ clean filters and coils;
- ▶ leakage reduction;
- ▶ energy recovery where clinically appropriate and permitted;
- ▶ occupancy-based turndown only where standards allow;
- ▶ BMS-based scheduling and fault detection.

Never

Do not reduce airflow in critical spaces merely to save energy without checking the applicable ventilation standard and risk assessment.

Numerical Example 3: Fan Energy Perspective

A supply fan moves 5 m³/s against a total pressure rise of 800 Pa. Assume overall fan-motor efficiency of 65%.

$$P \approx \frac{Q\Delta P}{\eta} = \frac{5 \times 800}{0.65} = 6154 \text{ W}$$

$$P \approx 6.15 \text{ kW}$$

Why It Matters

A dirty filter that increases system pressure drop can increase fan energy or reduce airflow, depending on the control strategy.

Case Study 1: OR Loses Positive Pressure

During surgery, the BMS alarms that the OR pressure has fallen to neutral.

Possible Causes

- ▶ supply fan problem;
- ▶ clogged filters;
- ▶ return/exhaust airflow too high;
- ▶ door left open;
- ▶ damper/control failure;
- ▶ nearby renovation changed air balance.

Immediate Question

Should surgery continue, pause or relocate? The answer requires coordination between the clinical team, infection control and engineering based on risk and institutional policy.

Case Study 2: All Room Alarm

A patient with suspected airborne infection is in an All room. The pressure monitor changes from negative to positive.

1. Keep the room door closed.
2. Notify clinical/infection-control staff.
3. Verify the pressure reading independently.
4. Check exhaust fan, supply airflow and door/envelope condition.
5. Apply contingency measures according to facility policy.
6. Do not return the room to service until required airflow direction is restored and documented.

Case Study 3: Condensation Above ICU Ceiling

Water droplets appear around a chilled-water pipe above an ICU ceiling.

- ▶ Is the insulation damaged or vapor barrier broken?
- ▶ Is room humidity unusually high?
- ▶ Is there a condensate drain leak?
- ▶ Could moisture affect electrical or medical-gas services?
- ▶ Is microbial contamination possible?

Lesson

Facility, infection-control and biomedical teams may all need to participate because a moisture problem can become an infrastructure, infection and equipment-safety problem.

Case Study 4: Imaging Room Overheats

A CT system shuts down repeatedly in the afternoon because the equipment room becomes too warm.

- ▶ verify HVAC capacity against equipment heat rejection;
- ▶ check whether recent equipment upgrades increased load;
- ▶ inspect airflow and filters;
- ▶ trend room temperature and chiller conditions;
- ▶ verify standby cooling strategy if 24/7 operation is required.

Root Cause Thinking

Repeated equipment shutdowns may be caused by facility HVAC, not by failure of the medical device itself.

Role of the Biomedical Engineer

Biomedical engineers should:

- ▶ communicate equipment environmental requirements to facility engineers;
- ▶ investigate failures potentially caused by temperature, humidity or airflow;
- ▶ coordinate commissioning of equipment rooms;
- ▶ ensure medical-device maintenance does not obstruct critical diffusers or returns;
- ▶ understand isolation/OR pressure implications during equipment service;
- ▶ participate in multidisciplinary risk assessments for new technology.

Role of the Hospital / Facility Engineer

Facility engineers are responsible for:

- ▶ maintaining HVAC availability and performance;
- ▶ managing chillers, AHUs, fans, pumps and controls;
- ▶ ensuring critical-room airflow and pressure;
- ▶ preventive maintenance and calibration;
- ▶ BMS alarm response;
- ▶ energy management;
- ▶ commissioning and renovation coordination;
- ▶ documentation and compliance with applicable standards.

Multidisciplinary HVAC Governance

Safe hospital HVAC needs collaboration among:

- ▶ facility/hospital engineering;
- ▶ biomedical/clinical engineering;
- ▶ infection prevention and control;
- ▶ clinicians and nursing staff;
- ▶ architects and MEP consultants;
- ▶ maintenance contractors;
- ▶ safety and quality teams;
- ▶ hospital administration.

Ventilation is an engineering system with clinical consequences.

Current Standards and Guidance

For design and operation, consult the standards adopted by the relevant authority having jurisdiction.

- ▶ **ANSI/ASHRAE/ASHE Standard 170-2025:** Ventilation of Health Care Facilities.
- ▶ **ASHRAE/ASHE Guideline 43-2025:** Operations Guideline for Ventilation of Health Care Facilities.
- ▶ **CDC Environmental Infection Control:** ventilation, AI/PE and engineering controls.
- ▶ Applicable Indian building, fire, infection-control and accreditation requirements.
- ▶ Equipment-manufacturer environmental specifications.

Professional Practice

Different authorities may adopt different editions. Always verify the legally applicable version for the project location.

Classroom Activity 1: Design an Isolation Room

A patient room is $6 \times 4 \times 3$ m.

1. Calculate the airflow required for 12 ACH.
2. Decide whether the room should be positive or negative for airborne isolation.
3. Draw the direction of airflow at the door.
4. Identify where you would place supply and exhaust grilles conceptually.
5. List three alarms/parameters you would monitor.

Work in groups for 8 minutes.

Classroom Activity 1: Quick Solution

Room volume:

$$V = 6 \times 4 \times 3 = 72 \text{ m}^3$$

For 12 ACH:

$$Q_h = 12 \times 72 = 864 \text{ m}^3/\text{h}$$

For airborne infection isolation:

- ▶ room should be negative relative to corridor;
- ▶ air should flow corridor $\rightarrow$ room;
- ▶ monitor pressure, temperature/RH and fan/exhaust status.

Classroom Activity 2: Hidden HVAC Failures

For each situation, identify the likely engineering issue:

1. OR temperature is correct but pressure is neutral.
2. All room shows negative pressure only when the door is closed very tightly.
3. HEPA filter has low differential pressure despite months of heavy use.
4. ICU ceiling has repeated fungal staining near a diffuser.
5. CT scanner trips only on hot afternoons.

Goal

Think beyond temperature and identify airflow, leakage, filtration, moisture and capacity problems.

Mini-Project Ideas

- ▶ HVAC risk audit of an ICU, OT or laboratory.
- ▶ Low-cost differential-pressure monitoring prototype for an isolation room.
- ▶ BMS dashboard for critical HVAC parameters.
- ▶ Energy analysis of hospital AHUs using filter pressure-drop data.
- ▶ Predictive-maintenance model for chiller or AHU faults.
- ▶ Computational or simplified airflow study of an operating room.
- ▶ Portable HEPA placement study for a simulated patient room.

Summary

- ▶ Hospital HVAC controls temperature, humidity, ventilation, filtration and pressure.
- ▶ ACH describes ventilation quantity, but airflow pattern also matters.
- ▶ Positive pressure protects clean spaces; negative pressure contains contaminants.
- ▶ HEPA filtration, room sealing and pressure monitoring are critical in selected spaces.
- ▶ OT, ICU, AII, PE, laboratories and technical rooms have different requirements.
- ▶ Maintenance, balancing, commissioning and BMS monitoring sustain design performance.
- ▶ HVAC reliability directly affects infection control, patient safety and equipment availability.

In a hospital, air is an engineered clinical resource.

Assignment

Select one hospital area: **OT, ICU, All room, radiology, laboratory or CSSD.**

Prepare a two-page HVAC engineering note containing:

1. clinical purpose of the area;
2. required pressure relationship and airflow direction;
3. major HVAC components;
4. required monitoring parameters;
5. two possible HVAC failure modes and their clinical consequences;
6. one energy-efficiency improvement that does not compromise safety.

Include

A simple airflow block diagram showing supply, room and return/exhaust path.

Suggested Reading

- ▶ ANSI/ASHRAE/ASHE Standard 170-2025, *Ventilation of Health Care Facilities*.
- ▶ ASHRAE/ASHE Guideline 43-2025, *Operations Guideline for Ventilation of Health Care Facilities*.
- ▶ CDC, *Guidelines for Environmental Infection Control in Health-Care Facilities*: Air section.
- ▶ G. D. Kunders, *Hospitals: Facilities Planning and Management*.
- ▶ B. M. Sakharkar, *Principles of Hospital Administration and Planning*.

Next Lecture: Hospital Water Supply, Plumbing and Sanitation Systems

Thank You

Questions and Discussion